Memory Protection

Lecture 9

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Memory Protection

- Memory Protection Unit
- Memory Management Unit

Memory Protection

ARM: MPU, RISC-V: PMP

Memory Management

MMU

Bibliography

for this section

1. **Andrew Tanenbaum**, *Modern Operating Systems (4th edition)*
   - Chapter 3 - *Memory Management*
     - Subchapter 3.3 - *Virtual Memory*
2. **Philipp Oppermann**, *Writing an OS in Rust*
   - *Introduction to Paging*
   - *Paging Implementation*

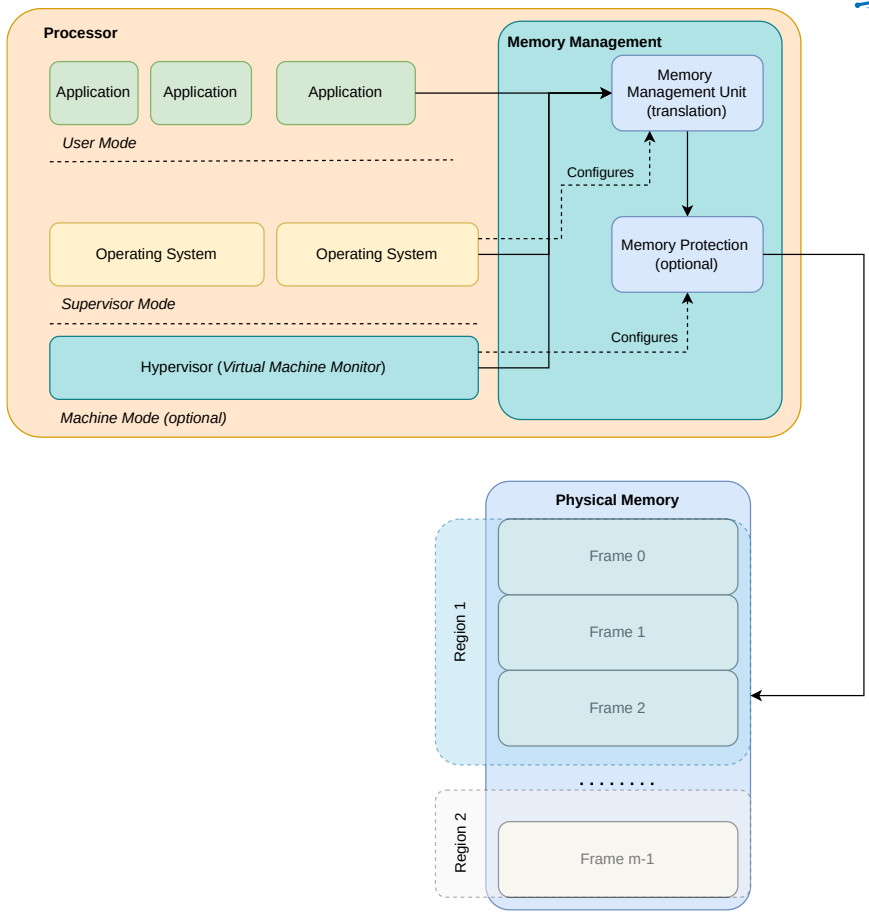
Memory Management

memory access defined page by page

- uses *logical addresses*
- **translates** to *physical addresses*

The processor works in at least two modes:

- **supervisor** mode
  - restricts access to some registers
  - accesses virtual addresses through Memory Protection (*if machine mode exists*)
- **user** mode
  - allows only ALU and memory load and store
  - accesses memory access through the Memory Management Unit (*MMU*)



Paging

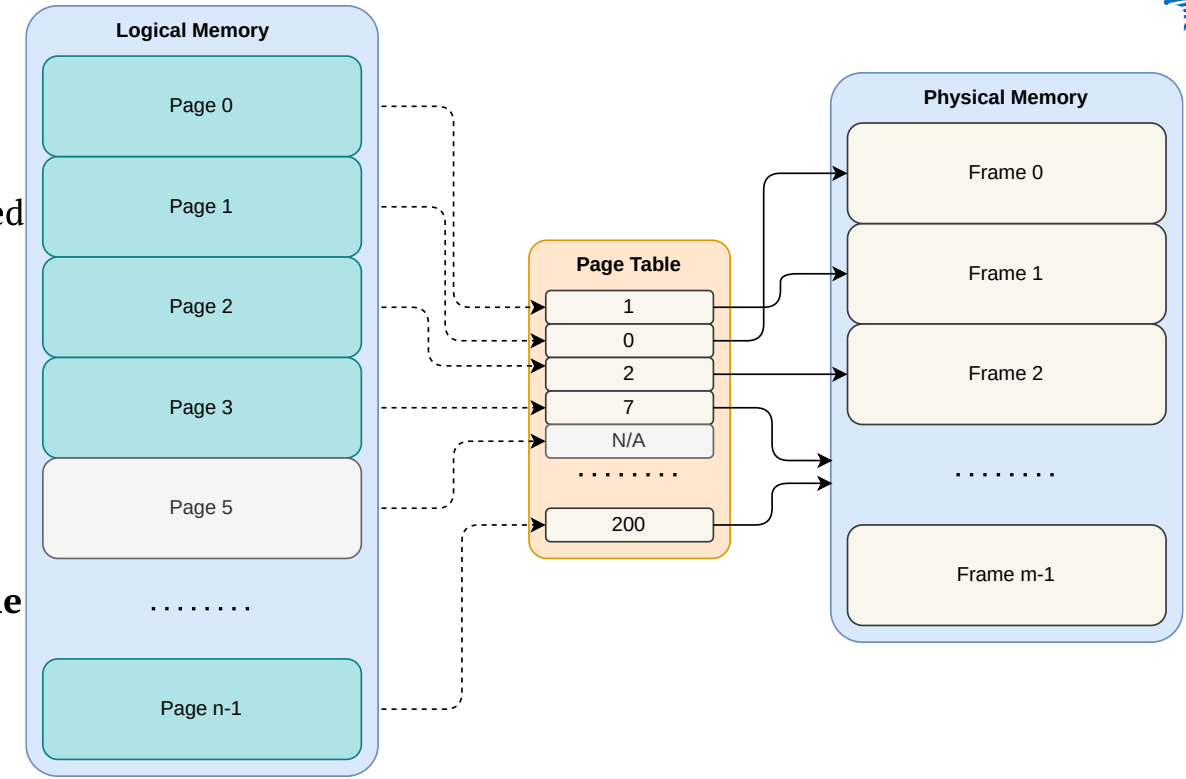
the memory *unit* is the page

- Physical Memory (*RAM*) is divided in **frames**
- Logical Memory is divided in **pages**
- *page* = *frame* = **4 KB** (usually)

logical addresses are translated to *physical addresses* using a **page table**

the **page table** is located in the **physical memory**

- each memory access requires at least 2 memory accesses



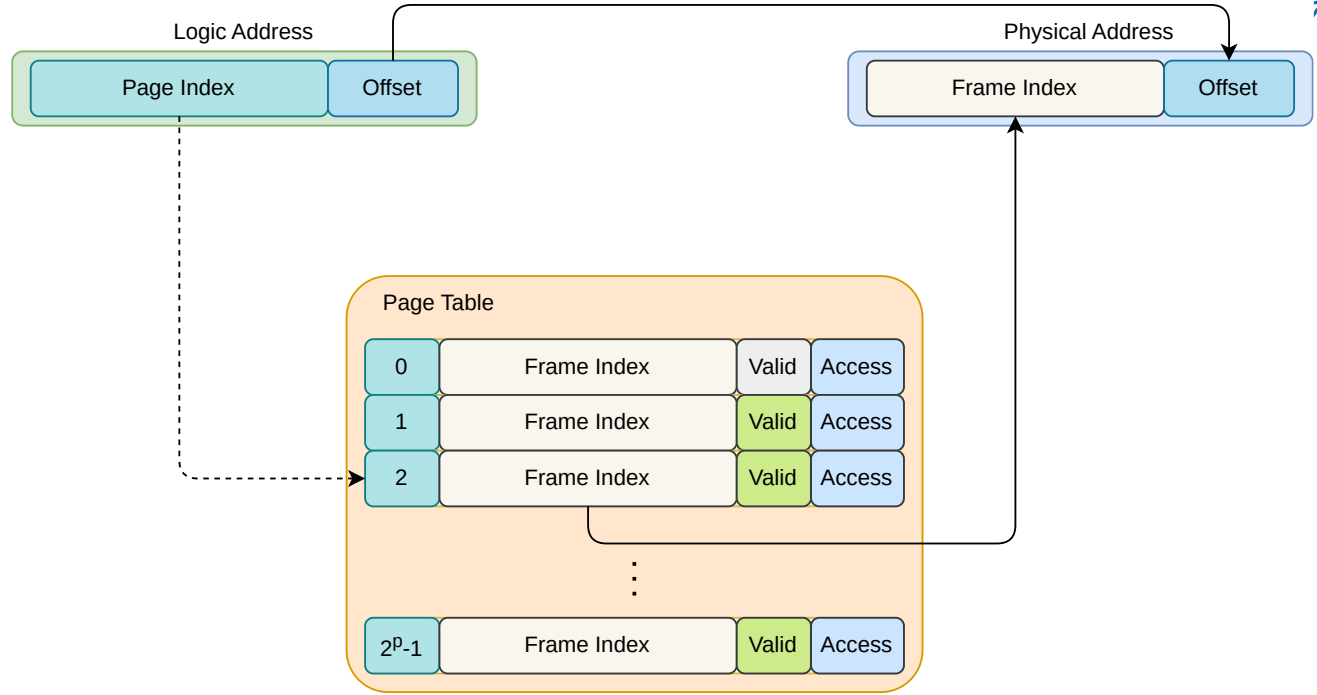
Address Translation

page to frame

the logic address is divided in two parts:

- *page index*
- *offset* within the page

the MMU translates every logic address into a physical address using a *page table*



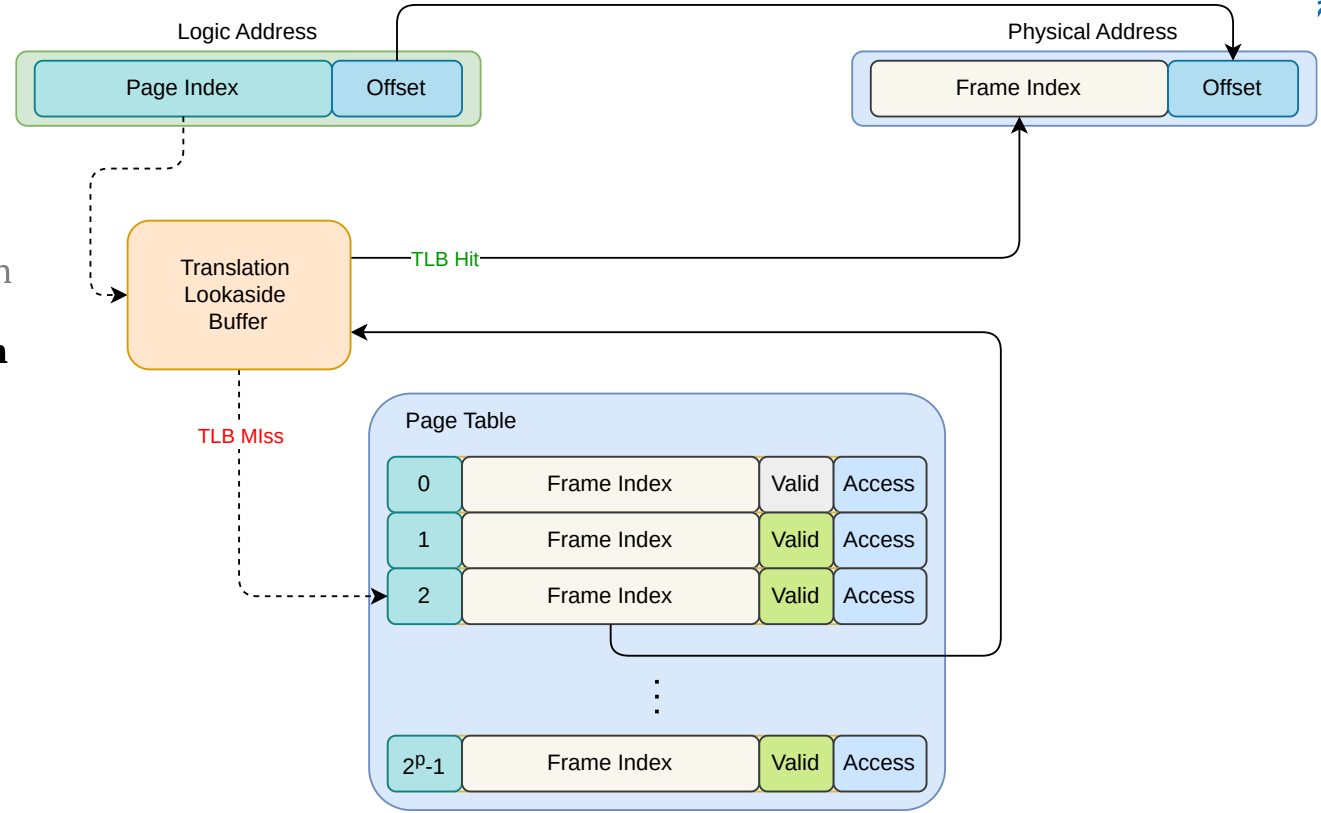
Translation Lookaside Buffer (TLB)

caching address translation

the **page table** is **stored in RAM**

each memory access **requires 2 accesses**

1. read the page table entry to translate the address
2. the requested access



Page Directory

caching address translation

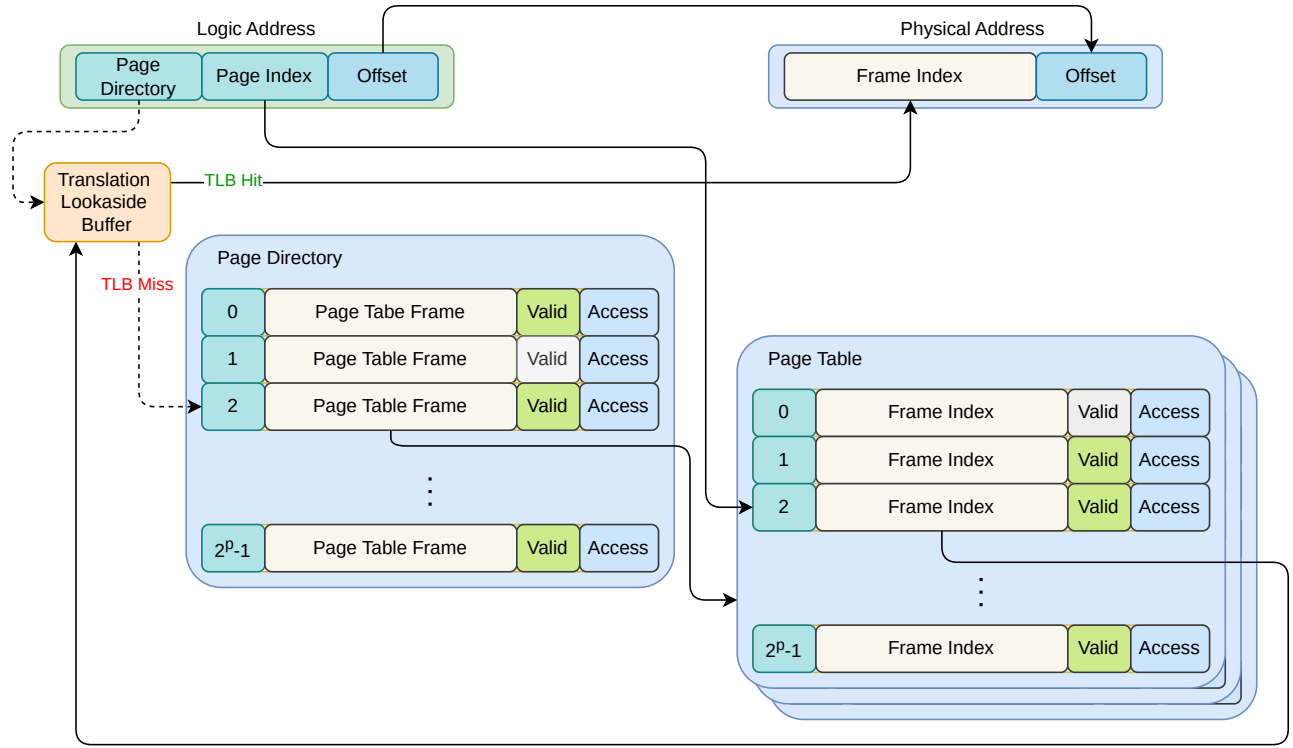
$$size_{table} = \frac{size_{ram}}{size_{page}}$$

- each table entry is 4B
- the address space is 4GB (for 32 bits processors)

$$size_{table_32_bits} = \frac{2^{32}}{4 \times 2^{10}}$$

$$size_{table_32_bits} = 4MB$$

RAM was counted in MB when paging started being used



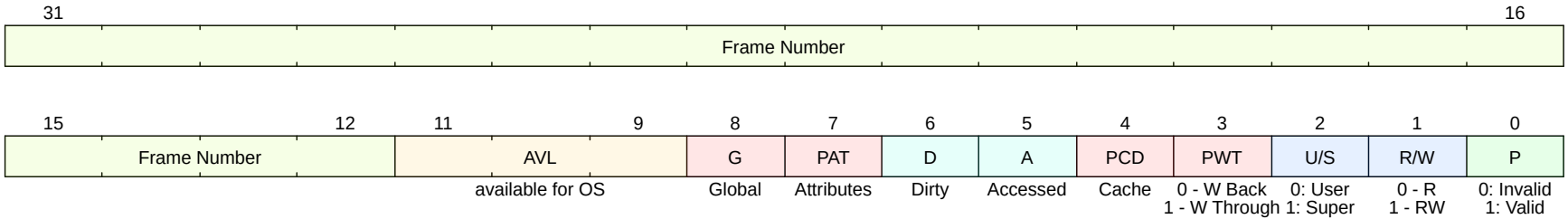
two levels, page directory and table, usually used for 32 bits systems

Page Table Entry

for x86 - 32 bits

this is one entry of the page table

- **P** - is the page's frame present in RAM?
- **R/W** - read only or read write access
- **U/S** - can the page be accessed in user mode?
- **D** and **A** - has this page been written since the OS has reset these bits?
- **AVL** - bits available for the OS to use, ignored by MMU

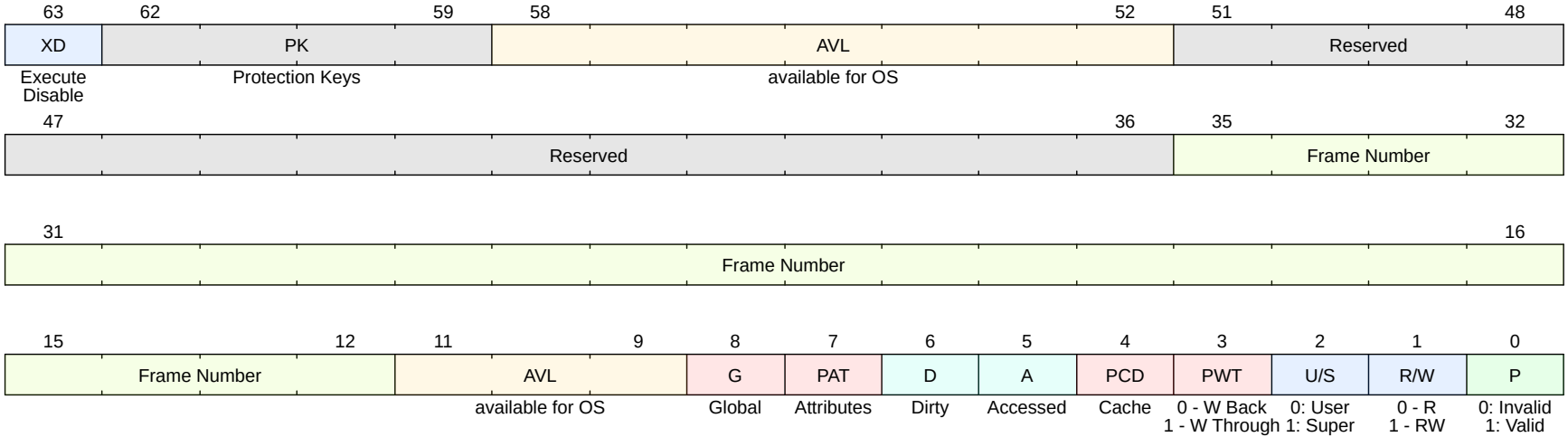


Page Table Entry

for x86 - 32 bits with PAE

this is one entry of the page table using Physical Address Extension (*PAE*)

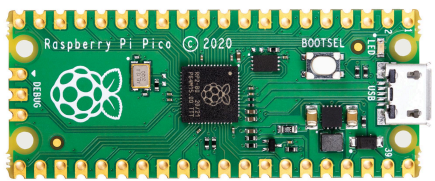
- **XD** - eXecute Disable (aka *DEP*), if set triggers a fault if an instruction is read from the page
- **PK** - Protection Keys, allows user mode to set protection (64 bit only)



Microcontroller (MCU)

Integrated in embedded systems for certain tasks

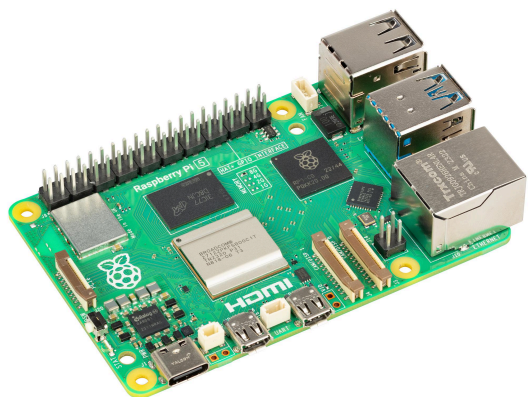
- low operating frequency (MHz)
- a lot of I/O ports
- controls hardware
- does not require an Operating System
- costs $0.1 - $25
- uses **Memory Protection Unit**



Microprocessor (CPU)

General purpose, for PC & workstations

- high operating frequency (GHz)
- limited number of I/O ports
- usually requires an Operating System
- costs $75 - $500
- uses **Memory Management Unit**



Conclusion

we talked about

- Memory Protection Unit
- Memory Management Unit